

Building Connected Internet of Things Widgets with Raspberry Pi and Intel Galileo



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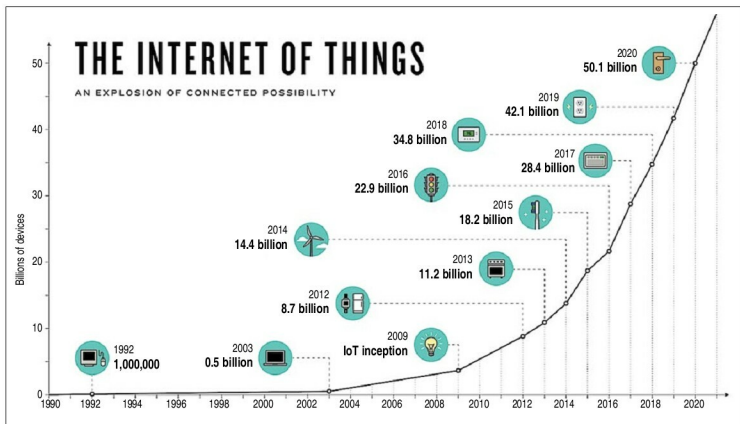
The Internet of Things (IoT), defined by the Oxford dictionary as a proposed development of the Internet in which everyday objects have network connectivity, allowing them to send and receive data, is the next big thing in embedded computing. Most of the information available on the Internet is made and added by human beings, but the IoT can change all this—data-collection by connected widgets can drastically change what we know about the physical world.

A boom owing to the arrival of the IoT is expected to connect a very large number of devices to the Internet. One major drawback was the limitation of 4.3 billion unique addresses posed by the IPv4 protocol. This issue was addressed with the development and subsequent adaptation of IPv6, which allows a nearly unlimited

number, 2^{128} , or approximately 3.4×10^{38} unique addresses. Gartner, the American research firm, predicts nearly 26 billion devices on the IoT by 2020.

The IoT can also be used to offshore data processing from small, low-power embedded microprocessors to large, powerful servers connected by the Internet. Another upside to connected widgets is that, large environmental trends, such as global warming, can be measured quantitatively by a large, global array of sensors. Blitzortung, a project dedicated to mapping and measuring lightning strikes around the world, makes use of such IoT sensors.

For such IoT devices, power usage and costs are primary concerns. Raspberry Pi, made by the Raspberry Pi Foundation, and Intel Galileo board, made by the silicon giant Intel, are especially suited for making



The IoT—an explosion of connected possibility